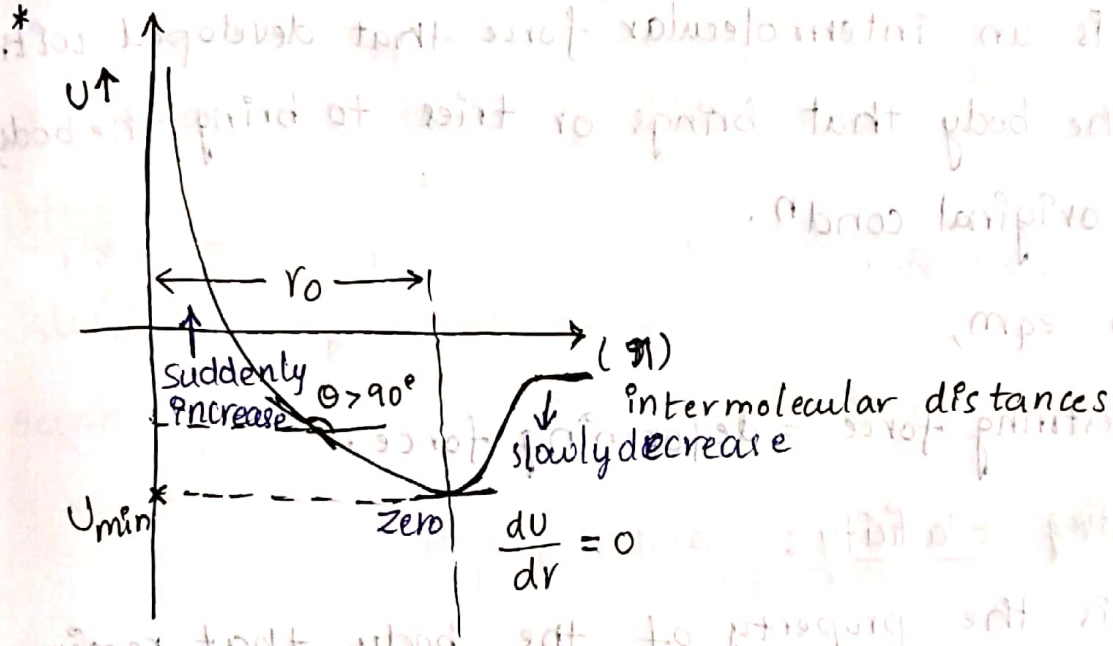
Mechanical Properties of Solids

Elasticity

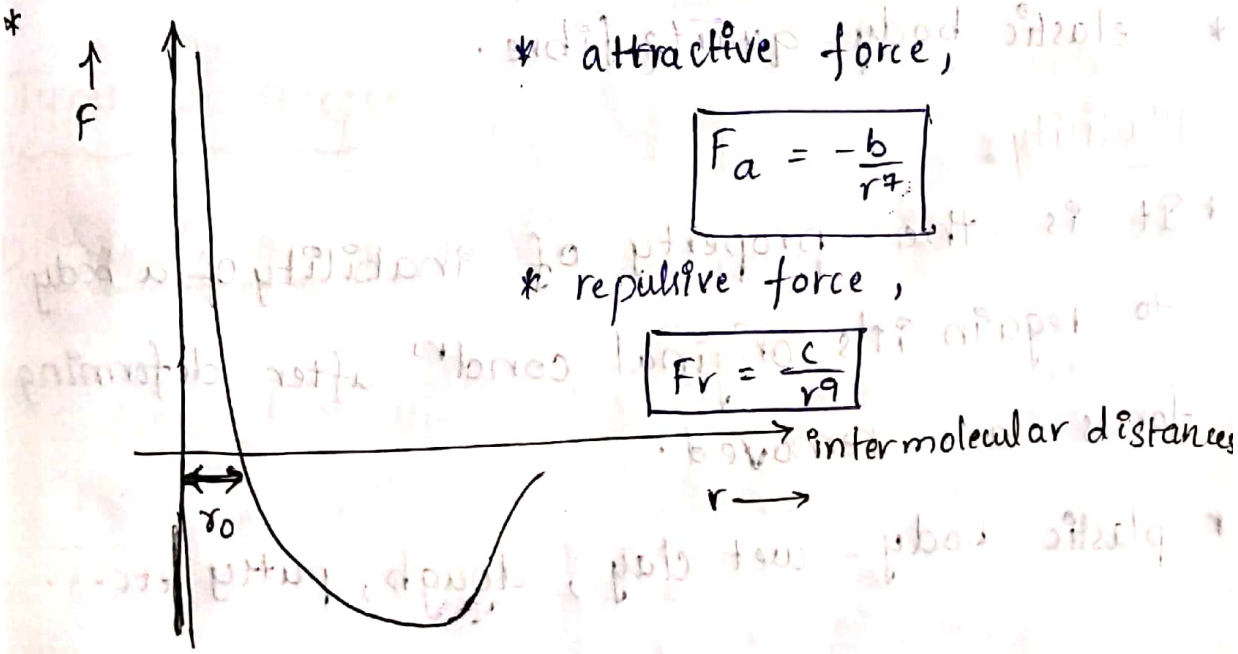
Intermolecular forces ⟶ "electrostatic forces".

attractive repulsive

* for any conservative force, $F = -\frac{dU}{dr}$



* atoms distance > r_0 — attractive
* atoms distance < r_0 — repulsive



* attractive force, $F_a = -\frac{b}{r^7}$
* repulsive force, $F_r = \frac{c}{r^9}$

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Deforming Force:

* It is an external force that tries to change the state of body.
  - [length, size, shape]

* diamond is perfectly rigid body.

Restoring force:

* It is an intermolecular force that developed with in the body that brings or tries to bring the body to original condn.

* In eqm,

  Restring force = deforming force.

Elasticity:

* It is the property of the body that regains its original condn after deforming forces are removed.

* elastic body - quartz fibre.

Plasticity:

* It is the property of inability of a body to regain its original condn after deforming forces are removed.

* plastic body - wet clay, dough, putty etc...

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Stress: (for elastic body)

* $\text{Stress} = \frac{\text{Restoring force}}{\text{Area}} = \frac{\text{Deforming force}}{\text{area}}$

⇒ units — N/m^2 or pascal — S.I

dyne/cm^2 — C.G.S

Strain:

$$\text{Strain} = \frac{\text{change in dimensions}}{\text{original dimesions}}$$

* It has no units & dimensions

Hooke's law:

With in the elastic limit.

According to hooke's, stress ∝ strain.

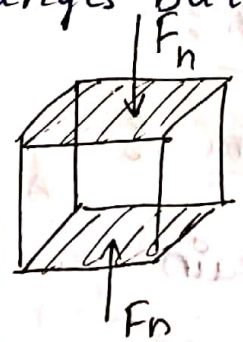
$$\text{stress} = \epsilon \times \text{strain}$$

$$\epsilon = \frac{\text{stress}}{\text{strain}}$$

ϵ — modulus of elasticity (elastic const)

Types of stress:

1) **Normal stress:** vol changes but shape doesnot

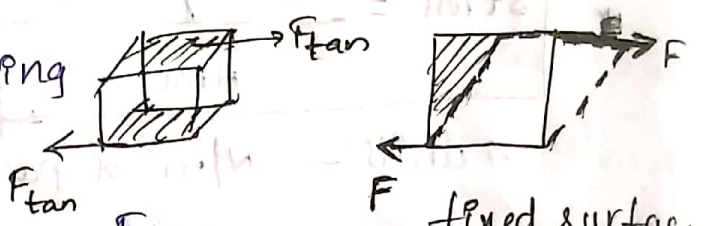


Normal stress $= \frac{F_n}{A}$

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2) Tangential Stress : shape changes but vol does not

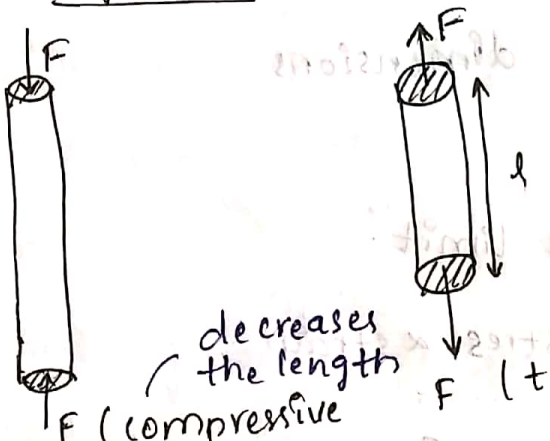
also called shearing stress.



Tangential stress = $\frac{F_{tan}}{A}$

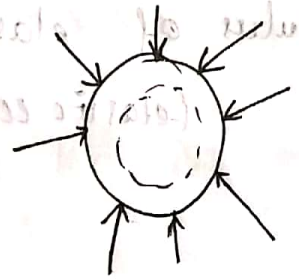
Types of Normal stress :

1) Longitudanal stress :



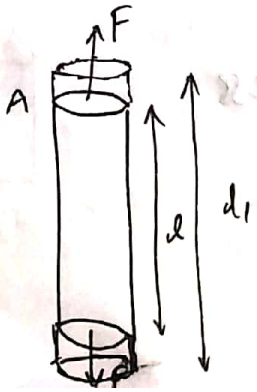
2) Volume | Bulk stress :

There will be change in volume.



Types of strain :

1) Longitudinal strain :



longitudinal strain

$= \frac{l_1 - l}{l} = \frac{\Delta l}{l} = \frac{e}{l}$

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e – elongation / extension in wire/rod.

for strain = 100%,

$$\frac{e}{l} \times 100 = 100$$

$$e = l$$

$$l_1 - l = l$$

$$\boxed{l_1 = 2l}$$

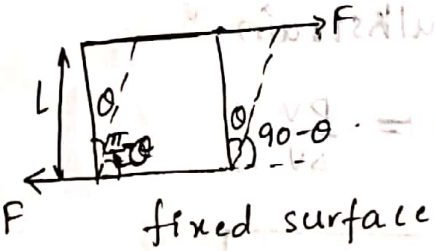
2) Bulk strain or vol strain :

$$\text{bulk strain} = \frac{V_1 - V}{V} = \frac{\Delta V}{V}$$



3) shearing strain :

Shearing strain is the amount of angle turned by a body.



Shearing strain = θ

from fig, $\tan\theta = \theta = \frac{\Delta x}{l}$

Types of Modulii of elasticity :

$$e = \frac{\text{stress}}{\text{strain}}$$

1) young's Modulus (Y) :

$$y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{\frac{F}{A}}{\frac{e}{l}} = \frac{Fl}{eA}$$

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$$Y = \frac{Fl}{eA}$$

* Y - depends on nature of material of wire & doesnot depend on its dimensions.
* for Y - large, large is stress & small is strain
* Eg: steel is more elastic than rubber bcoz Y is more for steel.
* Y - is more, Elasticity is more.

2) Bulk Modulus (B or K):

$$B = \frac{\text{Bulk stress}}{\text{Bulk strain}}$$

$$B = \frac{P}{\frac{\Delta V}{V}} = \frac{PV}{\Delta V}$$

In general, $B = \frac{-PV}{\Delta V}$

compressibility,

$$C = \frac{1}{B}$$

* B is more, Compressibility is less.

3) Rigidity Modulus (η):

$$\eta = \frac{\text{Shearing stress}}{\text{Shearing strain}}$$

$$\eta = \frac{F/A}{\theta}$$

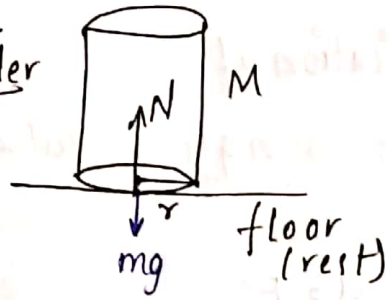
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$$\eta = \frac{F}{A\theta} = \frac{FL}{A\Delta x}$$

* η is large, twisting is less ($\theta \downarrow$).

Examples:

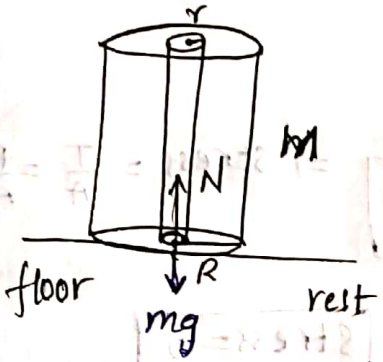
① solid cylinder



$N = mg$

$$\text{Stress} = \frac{Mg}{A} = \frac{Mg}{\pi r^2}$$

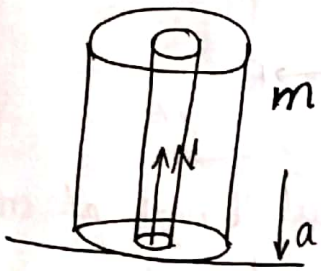
②



$N = mg$

$$\text{Stress} = \frac{mg}{A} = \frac{mg}{\pi(R^2 - r^2)}$$

③



Here $ma = mg - N$

$N = m(g - a)$

$$\text{Stress} = \frac{N}{A} = \frac{m(g-a)}{\pi(R^2 - r^2)}$$

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If $a = g$, $\boxed{\text{Stress} = 0}$

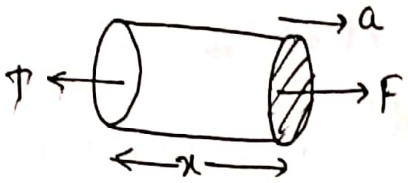
Eg: Find



① stress as a func^{l} of x.

② Total elongation in the wire due to the application of this force. (Y – young's modulus)

Sol^{n}:



$$a = \frac{F}{M}$$

$$F - T = ma$$

$$F - T = \frac{m}{L} x \frac{F}{m}$$

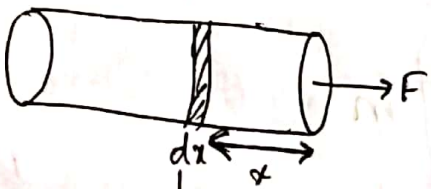
$$\boxed{T = F\left(1 - \frac{x}{L}\right)} \Rightarrow \text{stress} = \frac{T}{A} = \frac{F}{A}\left(1 - \frac{x}{l}\right)$$

at free end, $x = L \Rightarrow \boxed{T = 0} \Rightarrow \boxed{\text{stress} = 0}$

at force end, $x = 0 \Rightarrow T = F$ & $\boxed{\text{stress} = \frac{F}{A}}$

To find total elongation,

$$Y = \frac{\text{stress}}{\text{strain}}$$



L → original length of small element.

$$\text{Strain} = \frac{\text{stress}}{Y}$$

$$\frac{de}{dx} = \frac{F}{AY}\left(1 - \frac{x}{L}\right)$$

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$$\int de = \int_0^L \frac{F}{Ay}\left(1-\frac{x}{L}\right)\cdot dx$$

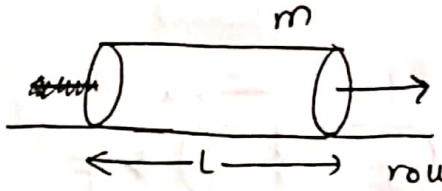
total elongation,

$$e = \frac{F}{Ay}\left[x - \frac{x^2}{2L}\right]_0^L$$

$$= \frac{F}{Ay}\left[L - \frac{L^2}{2L}\right]$$

$$\boxed{e = \frac{FL}{2Ay}}$$

Eg:



Find Total elongation in the rod if

① $F < \mu mg$ ② $F > \mu mg$.

Sol:

① $F < \mu mg$

wire/rod is at rest.

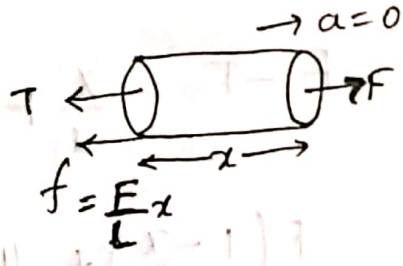
i.e, $\boxed{a = 0}$



$T + f = F$

$T + \frac{F}{L}(x) = F$

$$\boxed{T = F\left(1 - \frac{x}{L}\right)}$$



$$\text{Stress} = \frac{T}{A} = \frac{F}{A}\left(1 - \frac{x}{L}\right)$$

now, $\text{strain} = \frac{\text{stress}}{Y}$

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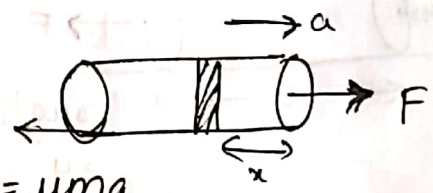
$$\frac{de}{dx} = \frac{F}{AY}\left(1 - \frac{x}{L}\right)$$

$$\int de = \frac{F}{AY}\int_0^L (1 - x/L)\cdot dx$$

$$e = \frac{F}{AY}\left(L - \frac{L}{2}\right).$$

$$\boxed{e = \frac{FL}{2AY}}$$

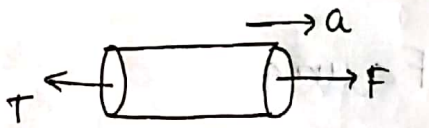
(9) $F > \mu mg$



$f = \mu mg$

$$F - \mu mg = ma$$

$$a = \frac{F - \mu mg}{m} \quad -(1)$$



$$F - T = m'a$$

$m' = \frac{m}{l}x$

$$F - T = \frac{m}{l}x\left(\frac{F - \mu mg}{m}\right) + \frac{\mu mg}{l}x$$

$$F - T = \frac{x}{l}F - \frac{\mu mgx}{l} + \frac{\mu mgx}{l}$$

$$F\left(1 - \frac{x}{l}\right) + \frac{\mu mgx}{l} = T + \frac{\mu mgx}{l}$$

$$\boxed{T = F\left(1 - \frac{x}{l}\right)}$$

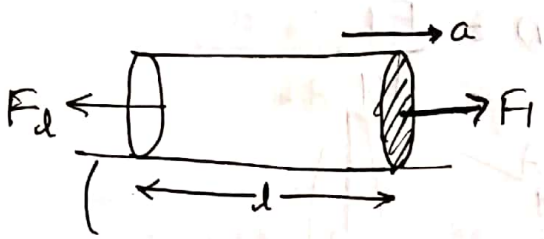
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$$\frac{de}{dx} = \frac{T}{AY}$$

$$de = \frac{T}{AY} \cdot dx$$

$$e = \frac{Fl}{2AY}$$

Eg:



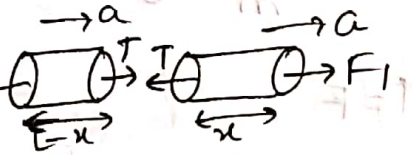
Smooth

$F_1 > F_2$

$$a = \frac{F_1 - F_2}{m} \quad -①$$



$$F_1 - T = \frac{m}{l} \times a$$



$$F_1 - T = \frac{m}{l} \times \left(\frac{F_1 - F_2}{m}\right)$$

$$T = F_1 - \frac{x}{l}(F_1 - F_2)$$

$$= F_1\left(1 - \frac{x}{l}\right) + F_2 \frac{x}{l}$$

$$\text{Stress} = \frac{T}{A} = \frac{F_1}{A}\left(1 - \frac{x}{l}\right) + \frac{F_2}{A}\frac{x}{l}$$

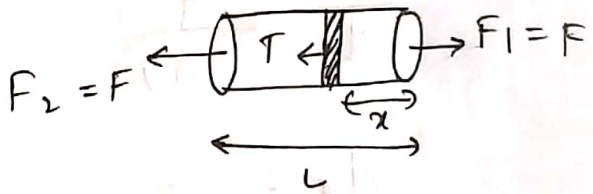
Total elongation,

$$de = \frac{F_1}{AY}\left(1 - \frac{x}{l}\right)dx + \frac{F_2 x}{AlY} \cdot dx$$

$$e = \frac{F_1}{A}\left(\frac{L}{2}\right) + \frac{F_2 L}{2} = \frac{(F_1 + F_2)L}{2AY}$$

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* If $F_1 = F_2$



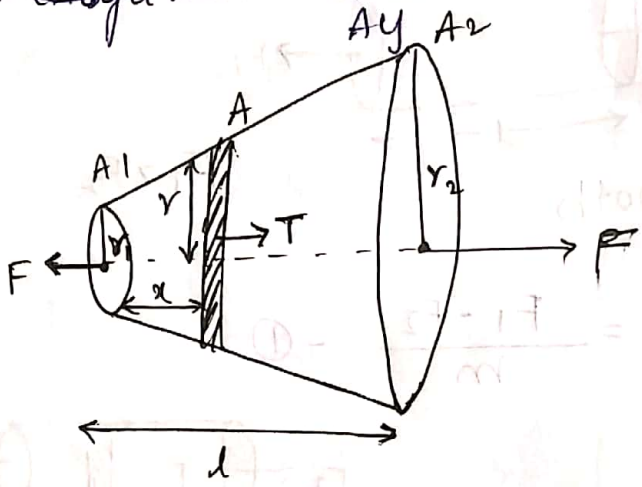
$\boxed{a = 0}$

① $T = F$

② strain $= \frac{T}{A} = \frac{F}{A}$ (const)

③ total elongation $= \frac{Fl}{AY}$

Eg:



① $a = \frac{F-F}{m} = 0$

② $T = F$

③ stress $= \frac{T}{A} = \frac{F}{A_1 + \left(\frac{A_2 - A_1}{l}\right)x}$

$l \rightarrow A_2 - A_1$

$1 \rightarrow \frac{A_2 - A_1}{l}$

$x \rightarrow \left(\frac{A_2 - A_1}{l}\right)x$

(If area is linearly increasing with x)

(or) stress $= \frac{T}{x} = \frac{T}{\pi R^2}$

where $r = r_1 + \frac{r_2 - r_1}{l}x$

(If radius is linearly increasing with x).

stress $= \frac{T}{\pi\left[r_1 + \frac{(r_2 - r_1)}{l}x\right]^2}$

now,

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$$\frac{de}{dx} = \frac{\text{stress}}{y}$$

$$e = \int \frac{F \cdot dx}{\left[A_1 + \frac{(A_1 - A_2)x}{l}\right] y}$$

$$e = \int \frac{Fl}{[A_1 l + (A_1 - A_2)x] y} \cdot dx$$

$$\boxed{e = \frac{Fl}{y(A_2 - A_1)} \ln \frac{A_2}{A_1}} \quad **$$

$$\boxed{e = \frac{Fl}{\pi r_1 r_2 y}} \quad **$$

$$\frac{de}{dx} \times y = \frac{F}{\pi \left[r_1 + \frac{(r_2 - r_1)x}{l}\right]^2}$$

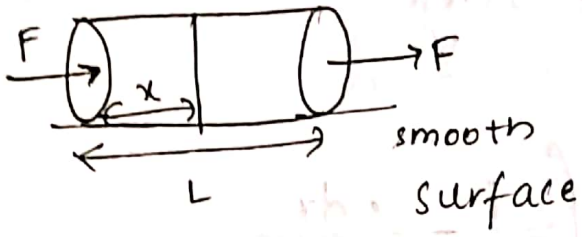
$$de = \frac{Fl^2}{\left[\pi \left[r_1 + (r_2 - r_1)x\right]^2\right] y} \cdot dx$$

$$e = \frac{Fl^2}{\pi (r_1 r_2) y (l^2)}$$

$$\boxed{e = \frac{Fl}{\pi r_1 r_2 y}}$$

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Ex:



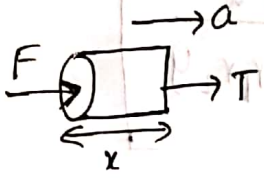
***

Find ① stress in the wire as a function of x.

② elongation in the wire.

③ displacement of C.M of wire w.r.to geometric centre of wire.

① $a = \frac{F+F}{m} = \frac{2F}{m}$



$F+T = \frac{m}{\ell} \times a$

$F+T = \frac{m}{\ell} \times \left(\frac{2F}{m}\right)$

$$\boxed{T = F\left(\frac{2x}{\ell} - 1\right)}$$

OK

$$\text{Stress} = \frac{T}{A} = \frac{F}{A}\left(\frac{2x}{\ell} - 1\right)$$

Points:

① for $x < \frac{\ell}{2} \Rightarrow T = -ve$

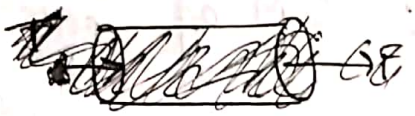
i.e, compressive stress in first half.



② for $x = \frac{\ell}{2} \Rightarrow T = 0 \Rightarrow$ stress $= 0$

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③ for $x > \frac{l}{2}$ ⇒ T = +ve



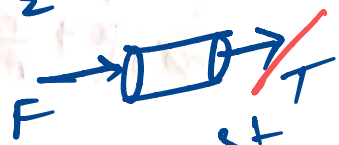
∴ stress in tensile.



$x > l/2$, $T \to +ve$ so assumed direction correct.

② now,

$$\frac{de}{dx} = \frac{T}{Ay}$$



1st half

$$\therefore de = \frac{F}{Ay}\left(\frac{2x}{l} - 1\right)$$



2nd half

$$e = \frac{F}{Ay}\left(\frac{2x^2}{2l} - x\right)_0^L$$

$$\boxed{e = 0}$$

∴ Final total length = Initial total length.

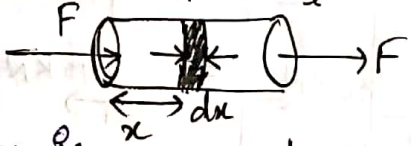
since mass density is "non uniform".

∴ C.M of rod is not coincides with geometric midpt of rod.

③ To find the Position of C.M of rod :

$dm = \frac{m}{l} \cdot dx$

location of CM of element of length dx is



here,

$dm = \lambda \cdot dx$

$= \lambda'(dx + dx)$



$$z = x + \int_0^x de$$

$$z = x + \frac{F}{Ay}\int_0^x \left(\frac{2x}{l} - 1\right) dx$$

$$z = x + \frac{F}{Ay}\left(\frac{x^2}{l} - x\right)$$

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∴ C.M of entire rod,

$$z_{cm} = \frac{\int z\,dm}{\int dm} = \frac{1}{m}\int\left[\bar{x} + \frac{F}{Ay}\left(\frac{x^2}{l} - x\right)\right] dm$$

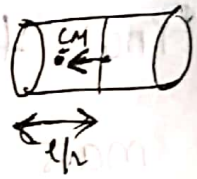
$$= \frac{1}{m}\int\left(x + \frac{F}{Ay}\left(\frac{x^2}{l} - x\right)\right)\cdot dx \cdot \frac{m}{l}$$

$$= \frac{1}{l}\int_0^l x\cdot dx + \frac{F}{Ayl}\int_0^l \left(\frac{x^2}{l} - x\right)\cdot dx$$

$$= \frac{l^2}{2l} + \frac{F}{Ayl}\left(\frac{2l^2}{3} - \frac{l^2}{2}\right)$$

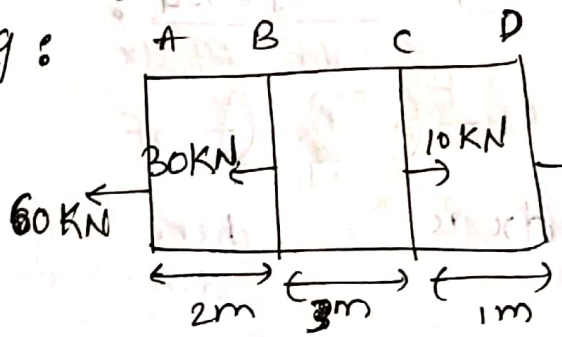
$$= \frac{l}{2} + \frac{F}{Ay}\left(\frac{-l}{6}\right)$$

$$= \frac{l}{2} - \frac{Fl}{6Ay}$$



∴ displacement $= \frac{Fl}{6Ay}$ towards left from geometric centre of the rod.

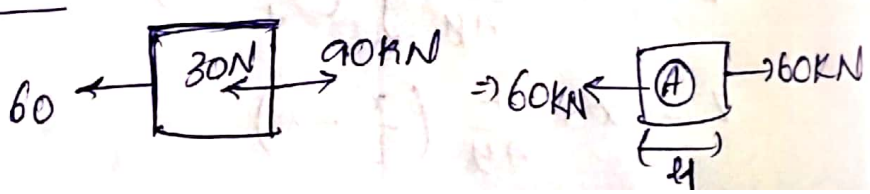
Eg:



Given $y = 2\times10^{11}\ N/m^2$

Find total elongation in rod.

for part AB:

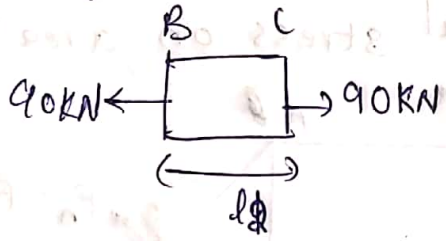


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$$e_1 = \frac{F_1 l_1}{AY} = \frac{60 \times 10^3 \times 2}{2 \times 10^{11} \times A}$$

$$\boxed{e_1 = \frac{6}{A} \times 10^{-7}}$$

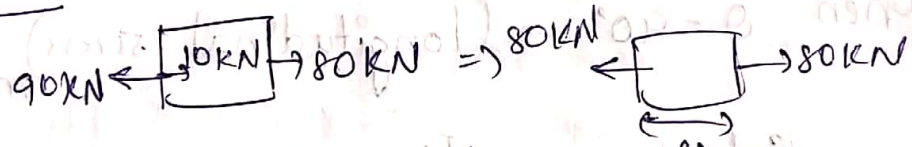
for part BC :



$$e_2 = \frac{F_2 l_2}{AY} = \frac{90 \times 10^3 \times 3}{A \times 2 \times 10^{11}}$$

$$\boxed{e_2 = \frac{135}{A} \times 10^{-8}}$$

for part CD :



$$e_3 = \frac{F_3 l_3}{AY} = \frac{80 \times 10^3 \times 1}{A \times 2 \times 10^{11}}$$

$$\boxed{e_3 = \frac{40}{A} \times 10^{-7}}$$

total elongation,

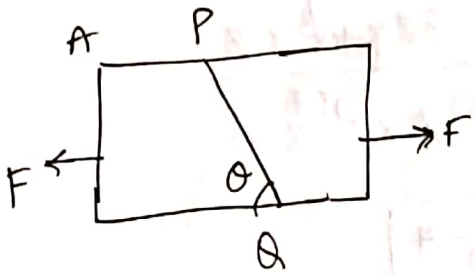
$$e = e_1 + e_2 + e_3$$

$$= \frac{(6 + 13.5 + 40) \times 10^{-7}}{A}$$

$$\boxed{e = \frac{23.5}{A} \times 10^{-7}}$$

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Eg:

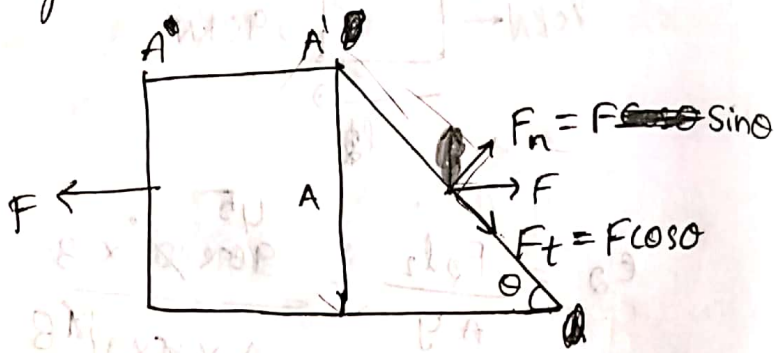


Find ① longitudinal stress.

② tangential stress on area PQ.

$\sin\theta = \frac{A}{A_1}$

$\Rightarrow A_1 = \frac{A}{\sin\theta}$



longitudinal stress $= \frac{F_n}{A} = \frac{F\sin\theta}{\frac{A}{\sin\theta}}$

$= \frac{F}{A}\sin^2\theta$

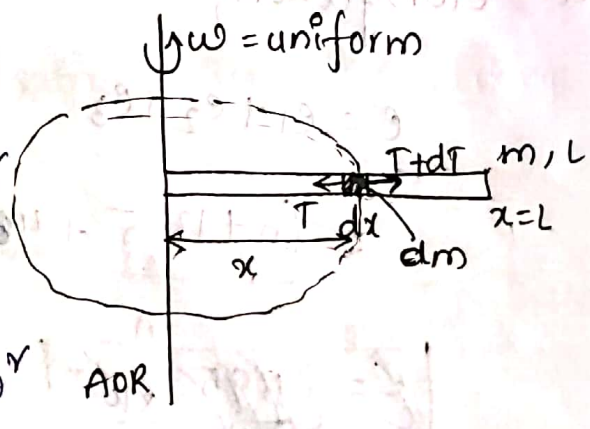
when $\theta = 90° \Rightarrow (\text{longitudinal stress})_{max} = \frac{F}{A}$

Tangential stress $= \frac{F_t}{A} = \frac{F\cos\theta}{\frac{A}{\sin\theta}} = \frac{F}{2A}\sin 2A$

when $2\theta = 90° \Rightarrow \theta = 45°$ $(\text{tang stress})_{max} = \frac{F}{2A}$.

Elongation in the rod due to rotation:

$F_{net} = (dm)(x\omega^2)$



$T - (T+dT) = (dm)x\omega^2$

$-dT = \frac{m}{L}dx \cdot x\omega^2$

$-dT = \frac{A\rho}{L}\cdot dx \cdot x\omega^2$ AOR

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$$-\int_{T}^{0} dT = A s \omega^2 \int_{x=x}^{x=L} x \cdot dx$$

$$\boxed{T = \frac{s A \omega^2 (L^2 - x^2)}{2}}$$

$$\text{stress} = \frac{T}{A} = \frac{1}{2} s \omega^2 (L^2 - x^2)$$

* At force end, $x = L \Rightarrow$ stress $= 0$

* At AOR, $x = 0 \Rightarrow$ stress $= \frac{1}{2} s \omega^2 L^2$ (max)

clearly stress $\propto \omega^2$

i.e, As $\omega \uparrow \Rightarrow$ stress $(\uparrow)$ and reaches to ruptured strength (σ).

$$\sigma = \frac{1}{2} s \omega_0^2 L^2$$

$$\boxed{\omega_0 = \frac{1}{L}\sqrt{\frac{2\sigma}{s}}} \quad **$$

* frequency of rotation,

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi L}\sqrt{\frac{2\sigma}{s}}$$

* To find elongation in rod :

$$\frac{de}{dx} = \frac{T}{Ay}$$

$$\frac{de}{dx} = \frac{1}{2} \frac{s \omega^2}{y} (L^2 - x^2) \cdot dx$$

∴ Total elongation,

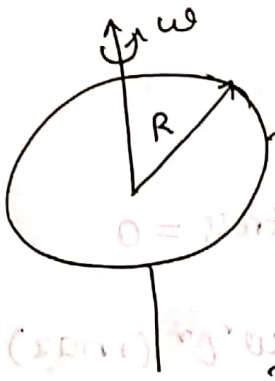
$$e = \int de = \int_{x=0}^{x=L} \frac{1}{2} \frac{s \omega^2}{y} (L^2 - x^2) \cdot dx$$

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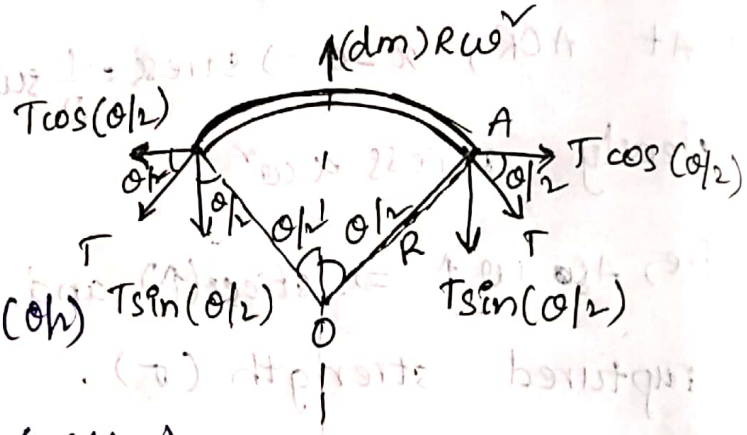
$$e = \frac{1}{2}\frac{\rho\omega^2}{4}\left[l^2 - \frac{l^2}{3}\right]$$

$$\boxed{e = \frac{\rho\omega^2 l^3}{24}}$$

Eg:



ring of mass m & radius 'R'



In eqm,

$$(dm)R\omega^2 = 2T\sin(\theta/2)$$

$$T\sin(\theta/2) \quad T\sin(\theta/2)$$

$$\frac{m}{2\pi R} \times R\theta \times R\omega^2 = 2T(\theta/2)$$

$$T = \frac{mR\omega^2}{2\pi} \qquad (m = A \times 2\pi R\rho)$$

$$T = \frac{A \times 2\pi R\rho \times mR\omega^2}{2\pi}$$

$$\boxed{T = \rho A R^2\omega^2}$$

* stress in the ring $= \frac{T}{A} = \rho\omega^2 R^2$

∴ when $\omega = \omega_0 \Rightarrow$ stress $= \sigma$ (breaking stress)

$$\boxed{\sigma = \rho\omega_0^2 R^2}$$

$$\Rightarrow \boxed{\omega_0 = \frac{1}{R}\sqrt{\frac{\sigma}{\rho}}}^{**}$$

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And $f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi R}\sqrt{\frac{\sigma}{\rho}}$

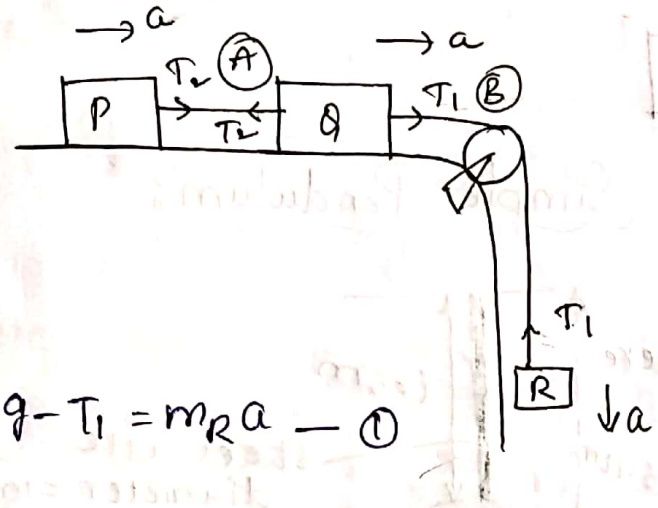
* strain = $\frac{\text{change in length}}{\text{original length}} = \frac{2\pi R_1 - 2\pi R}{2\pi R} = \frac{\Delta R}{R}$

we have, strain = $\frac{\text{stress}}{Y}$

$$\frac{\Delta R}{R} = \frac{\rho \omega^2 R^2}{Y}$$

Increase in radius $(\Delta R) = \frac{\rho \omega^2 R^3}{Y}$

Eg:



Find longitudinal strains in strings A & B.

$m_R g - T_1 = m_R a$ — ①

$T_1 - T_2 = m_Q a$ — ②

$T_2 = m_P a$ — ③

solving eqns ①, ②, ③

we get T_1, T_2, a :

now, stress in wire A = $\frac{T_2}{A_A}$

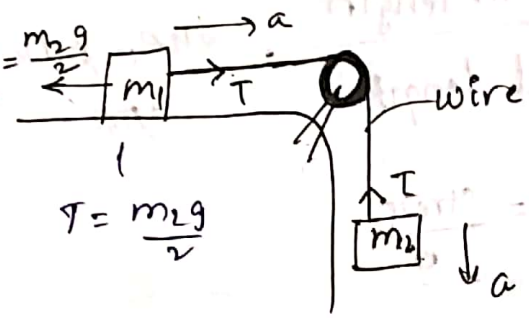
strain in wire A = $\frac{T_2}{A_A Y_A}$

dly stress in wire B = $\frac{T_1}{A_B}$

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Strain in wire B = $\frac{T_1}{a_B \, y_B}$

Ex: $F = \frac{m_2 g}{2}$



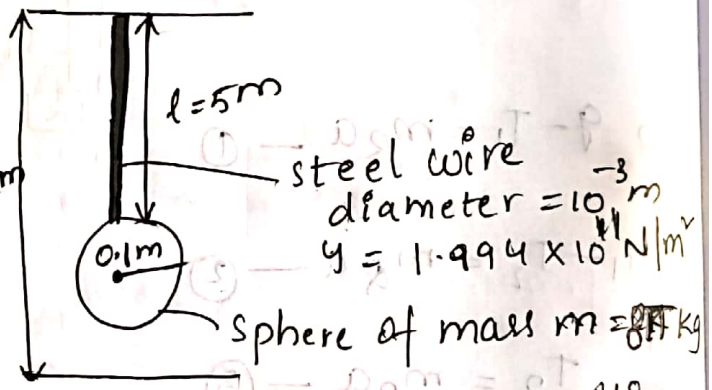
$$m_2 g - T = m_2 a \Rightarrow T = m_2(g-a)$$

$$\Rightarrow \frac{m_2 g}{2} = m_2 g - m_2 a$$

$$\boxed{a = g/2}$$

Oscillating in Simple Pendulum:

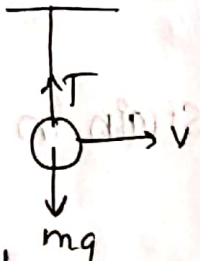
Eg: where the sphere is made to oscillate as a simple pendulum. It just grazes the floor at bottom most position. Find speed of sphere at this position.



elongation in wire = 5.22 − 5.2 = 0.02m.

$$T - mg = \frac{mv^2}{r}$$

$$T = mg + \frac{mv^2}{r}$$



$r = 5.22 - 0.1$

$= 5.12$m

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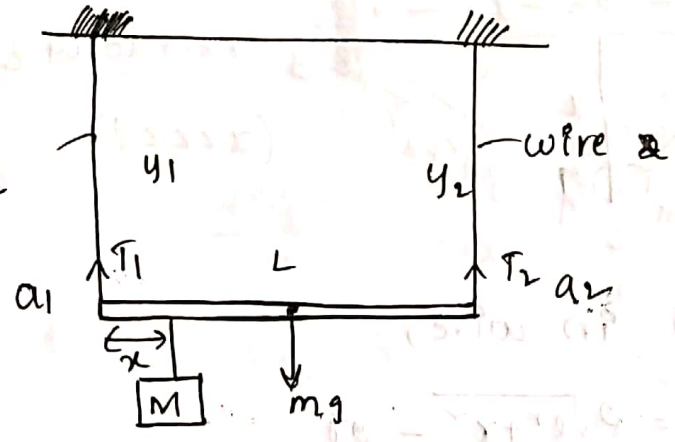
now, $e = \frac{T_1}{Ay}$, $l = 5m$.

$$T = \frac{yAe}{l}$$

$$mg + \frac{mv^2}{r} = \frac{yAe}{l}$$

solving we get $V = 8.8 m/s$.

Eg 8



where should mass be placed (hanged) from left end so that,

① stress in both the wires are equal

② strain in both the wires are equal.

In eqm,

$$T_1 + T_2 = (m+M)g \quad \text{— ①}$$

Turning torque about left end of rod is zero.

$$mgx + mg\frac{l}{2} = T_2 L \quad \text{— ②}$$

① Given stress in both the wires are equal i.e, $\frac{T_1}{a_1} = \frac{T_2}{a_2}$ — ③

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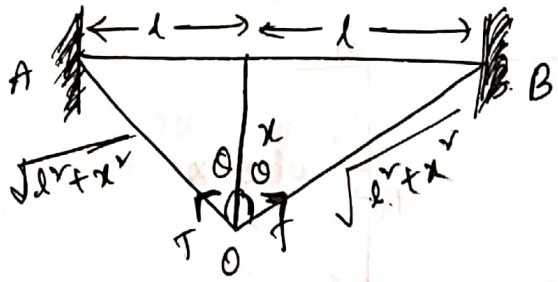
solving we get T_1, T_2, x.

(2) given strain in both wires are equal.

$$\frac{T_1}{a_1 y_1} = \frac{T_2}{a_2 y_2} \quad -(4)$$

solving we get T_1, T_2, x.

Eg:



r = radius of wire

$(x \ll l)$

* elongation in wire,

$$e = 2\sqrt{l^2+x^2} - 2l$$

$$e = 2l\sqrt{1+\left(\frac{x}{l}\right)^2} - 2l$$

$$e = 2l\left[\sqrt{1+\left(\frac{x}{l}\right)^2} - 1\right]$$

$$e = 2l\left(1 + \frac{x^2}{2l^2} - 1\right)$$

$$\boxed{e = \frac{x^2}{l}}$$

* Tension in wire,

$$T = \frac{yAe}{2l}$$

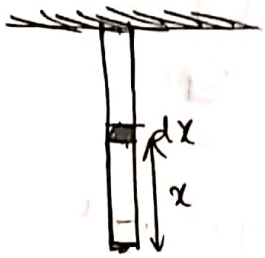
$$\boxed{T = \frac{y\pi r^2 x^2}{2l^2}}$$

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Elongation in the wire due to its own weight :

mass of wire

$\Rightarrow m = Al\rho$



Force on small element

= wt of wire below to it

$$T = \frac{m}{l} x g = \frac{Al\rho x g}{l} = A\rho x g$$

$$\text{stress} = \frac{T}{A} = \frac{A\rho g x}{A} = \rho g x$$

we have,

$$\frac{de}{dx} = \frac{\rho g x}{y}$$

$$de = \frac{\rho g x}{y} dx$$

$$e = \frac{\rho g}{y} \int_{x=0}^{l} x \cdot dx$$

$$\boxed{e = \frac{\rho g l^2}{2y}}$$

$e = \frac{Fl}{Ay}$

$e = \frac{(0+mg)\cdot l}{2Ay}$

$= \frac{mgl}{2Ay}$

$= \frac{\rho g l^2}{2y}$

* The max length of wire that can be hanged without breaking due to its own wight is

$$l = \frac{\text{breaking stress}}{\rho g}$$

Shortcut -

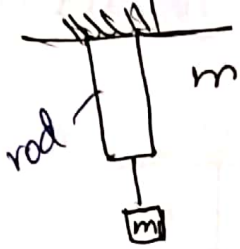
① 

$$e = \frac{F(l/2)}{Ay}$$

$$e = \frac{Fl}{2Ay} = \frac{mgl}{2Ay}$$

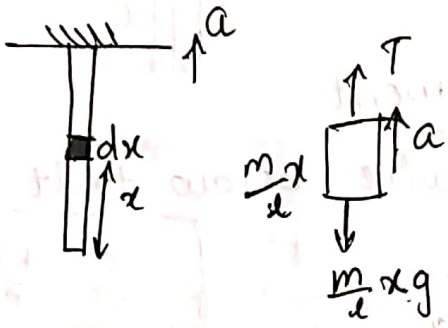
$$\boxed{e = \frac{\rho g l^2}{2Ay}}$$

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(2) 

elongation,

$$e = e_1 + e_2 = \frac{mgl}{2Ay} + \frac{mgl}{Ay}$$

(3) 

$$T - \frac{m}{l}xg = \frac{m}{l}xa$$

$$T = \frac{m}{l}x(g+a)$$

$$T = A\rho x(g+a)$$

$$\frac{T}{A} = \frac{\rho x(g+a)}{y}$$

$$\frac{de}{dx} = \frac{\rho x(g+a)}{y}$$

$$de = \frac{\rho(g+a)}{y} \cdot x \cdot dx$$

$$\boxed{e = \frac{\rho(g+a)l^2}{2y}}$$

Imp formulaes:

* Young's Modulus of wire:

$$y = \frac{Fl}{Ae} \Rightarrow \frac{1}{e} = \frac{Ay}{Fl} \Rightarrow \boxed{e = \frac{Fl}{Ay}}$$

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If l, A, y are const. then $e \propto F$.

Eg

*** l_1 is the length of wire. when force is F_1 and length is l_2 when force is F_2.

Find ① original length of wire

② length. l_3 when applied force is F_3.

$e \propto F$, l = original length of wire.

$l_1 - l \propto F_1$ — ①

$l_2 - l \propto F_2$ — ②

$l_3 - l \propto F_3$ — ③

$\frac{①}{②}$ $\quad \frac{l_1 - l}{l_2 - l} = \frac{F_1}{F_2}$

$l_1 F_2 - l F_2 = l_2 F_1 - F_1 l$

$$\boxed{\frac{l_1 F_2 - l_2 F_1}{F_2 - F_1} = l}$$

Again,

$\frac{①}{③}$ $\quad \frac{l_1 - l}{l_3 - l} = \frac{F_1}{F_3}$

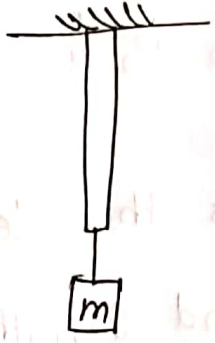
we get l_3.

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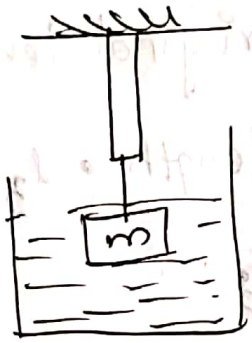
Ex:

$e \propto F$

$\Rightarrow e_1 \propto mg$



$e_2 \propto mg\left(1-\frac{dl}{d_B}\right)$



$$\frac{e_1}{e_2} = \frac{1}{\left(1-\frac{dl}{d_B}\right)}$$

(2) V = vol of the wire $= Al$

we have,

$$e = \frac{Fl}{Ay}$$

(1) $e = \frac{FV}{A^2 y}$ (2) $e = \frac{Fl^2}{Vy}$

or $\boxed{e = \frac{Fm}{\rho A^2 y}}$ $\boxed{e = \frac{Fl^2\rho}{my}}$

(3) Combination of wires:

1) Series Combination -

here force on each wire is same

i.e, $F_1 = F_2 = mg$.

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Here

total elongation,



$$e = e_1 + e_2 = \frac{F l_1}{A y_1} + \frac{F l_2}{A y_2}$$

$$e = \frac{F}{A}\left(\frac{l_1 y_2 + l_2 y_1}{y_1 y_2}\right)$$

or

$$\boxed{e = \frac{F}{A}\frac{(l_1 + l_2)}{y_s}}$$

$$\boxed{\frac{l_1 + l_2}{y_s} = \frac{l_1}{y_1} + \frac{l_2}{y_2}}$$

In general,

$$\frac{l_1 + l_2 + l_3 + \cdots}{y_s} = \frac{l_1}{y_1} + \frac{l_2}{y_2} + \frac{l_3}{y_3} + \cdots$$

Combine this with springs Mohith tyagi (CAJ SOR)

$K_x = \frac{YAe}{l}$

$K_x = \left(\frac{YA}{l}\right) e$

$e = \frac{Fl}{AY}$

* If wires have diff area of crossections.

$$\frac{l_1}{A_1 y_1} + \frac{l_2}{A_2 y} + \cdots = \frac{l_1 + l_2 + \cdots}{A y_s}$$

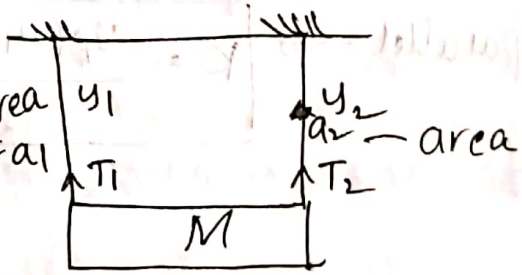
$$\left(\frac{l_1}{A_1} + \frac{l_2}{A_2} + \cdots\right)\frac{1}{y_s}$$

2) Parallel combination

here

① elongation in both the wires is same.

② stress in both wires need not to be same.



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$$T_1 + T_2 = mg \quad -(1)$$

& $e_1 = e_2$

$$\frac{T_1 \cancel{l}}{a_1 y_1} = \frac{T_2 \cancel{l}}{a_2 y_2}$$

$$\frac{T_1}{a_1 y_1} = \frac{T_2}{a_2 y_2} \quad -(2)$$

and $$mg = \frac{y_1 a_1 e}{l} + \frac{y_2 a_2 e}{l}$$

$$\frac{y_p (a_1 + a_2) \cancel{l}}{\cancel{l}} = (y_1 a_1 + y_2 a_2) \frac{\cancel{e}}{\cancel{l}}$$

$$\boxed{y_p = \frac{y_1 a_1 + a_2 y_2}{a_1 + a_2}}$$

Force Constants

$$F = \frac{yAe}{l}$$

$$F = ke$$

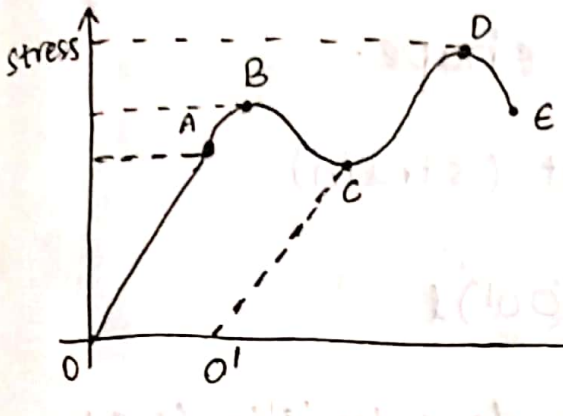
$$k = \frac{yA}{l} = \text{force const}$$

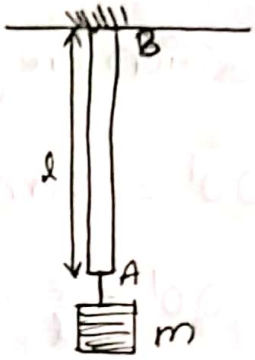
Series → $$\boxed{k = \frac{y_s A}{l_1 + l_2}}$$

parallel → $$\boxed{k = \frac{y_p (A_1 + A_2)}{l}}$$

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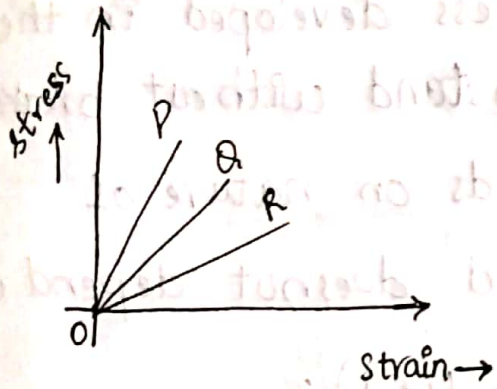
Behaviour of a wire under gradually Increasing Load:



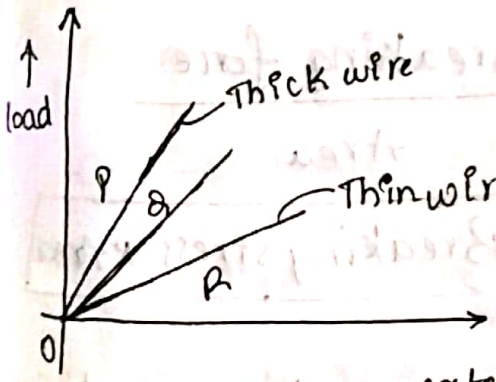


OA → linear part of graph.

Slope of line OA = young's modulus of wire.



$Y_P > Y_Q > Y_R$



$A_P > A_Q > A_R$

$Y = \frac{Fl}{Ae}$

$\frac{F}{e} = \frac{YA}{l}$ [for Y, l are const]

slope $= \frac{F}{e} \propto A$

[A] = proportionality limit :- Maximum value of stress upto which wire obeys hooke's law.

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B = elastic limit :- Max value of stress upto which wire comes in to it's original position / regains it's shape.

OO' = permendnt set (strain)

* $OO' = \frac{e}{l} \Rightarrow e = (OO')l$

C = yeild point : it elongates like fluid.

D = ultimate tensile stress (or) breaking stress :-

Max value of stress developed in the wire which can withstand without breaking

* breaking stress depends on nature of material of wire and doesnot depend on it's dimensions (length, area).

$$\text{Breaking stress} = \frac{\text{Breaking force}}{\text{Area}}$$

$$\boxed{\text{Breaking force} = \text{Breaking stress} \times \text{Area}}$$

Breaking force $\propto$ Area & doesnot depend on it's length.

Eg: the maximum length of wire required without breaking due to it's own wt.

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$$\text{B·stress} = \frac{mg}{A} = \frac{A l \rho g}{A}$$

$$\boxed{l = \frac{\text{B·stress}}{\rho g}} \quad **$$

* If wire is dipped in liquid,

$$\text{B·stress} = \frac{mg\left(1 - \frac{d_l}{d_b}\right)}{A}$$

$$\text{B·stress} = \frac{A \rho_b g l \left(1 - \frac{d_l}{d_b}\right)}{A}$$

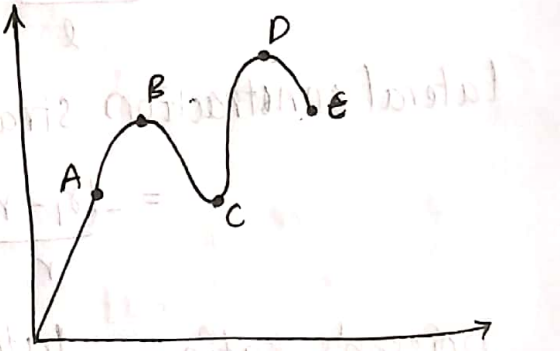
$$\text{B·stress} = g l \left(d_b - d_l\right)$$

$$\boxed{l = \frac{\text{B·stress}}{g(d_b - d_l)}}$$

* From graph,

OAB → elastic region

BCDE → plastic region



*If length of BCDE is more, then those materials are used to prepare thin wires called "ductile materials".

Eg: Gold, Copper, Al, Fe etc...

*If length of BCDE is less then the

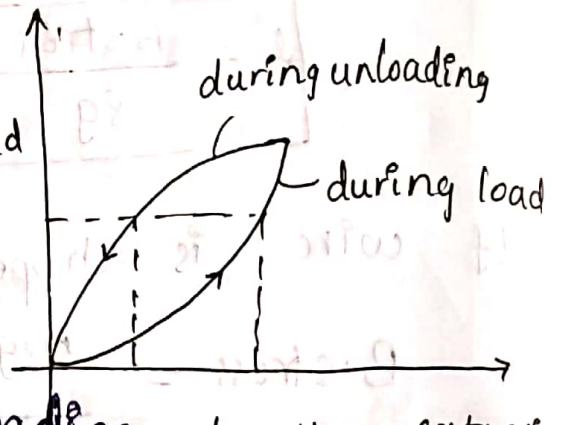
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materials are called "brittle materials".

Eg: Glass.

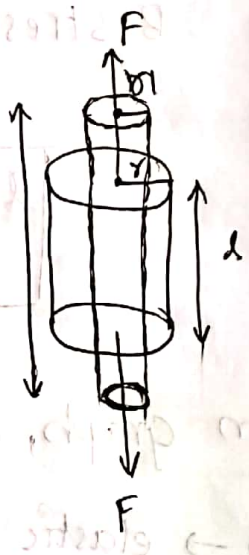
Hysterisis:

* The lagging of strain developed in the body during unloading than loading at the same load is called "hysterisis".



* Area of hysterisis curve = heat energy produced.

Poisson's Ratio (σ):



longitudinal strain

$$= \frac{l_1 - l}{l} = \frac{\Delta l}{l}$$

lateral contraction strain

$$= -\frac{(r_1 - r)}{r} = \frac{-\Delta r}{r}$$

$$\text{poisson's ratio} = \frac{\text{lateral contraction strain}}{\text{longitudinal strain}}$$

$$= \frac{-\Delta r}{r} \times \frac{l}{\Delta l}$$

$$\boxed{\sigma = \frac{-l \Delta r}{r \Delta l}}$$

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It has no units & dimensions.

theoritical limits → -1 to 0.5

practical limits → 0 to 0.5

we have,

volume of the wire,

$$V = \pi r^2 l$$

on differentiating,

$$\Delta V = \pi r^2 \Delta l + \pi l (2r \Delta r)$$

now

$$\frac{\Delta V}{V} = \frac{\pi r^2 \Delta l}{\pi r^2 l} + \frac{\pi l (2r \Delta r)}{\pi r^2 l}$$

$$\boxed{\frac{\Delta V}{V} = \frac{\Delta l}{l} + 2\frac{\Delta r}{r}}$$

$$\frac{\Delta V}{V} = \frac{\Delta l}{l} \Rightarrow 2\sigma \cdot \frac{\Delta l}{l}$$

$$\boxed{\frac{\Delta V}{V} = \frac{\Delta l}{l}(1-2\sigma)}$$

$\frac{\Delta V}{V}$: Volume Strain

$\frac{\Delta l}{l}$: linear strain

If $\frac{\Delta V}{V} = 0 \Rightarrow 1 - 2\sigma = 0$

$$\boxed{\sigma = 1/2}$$

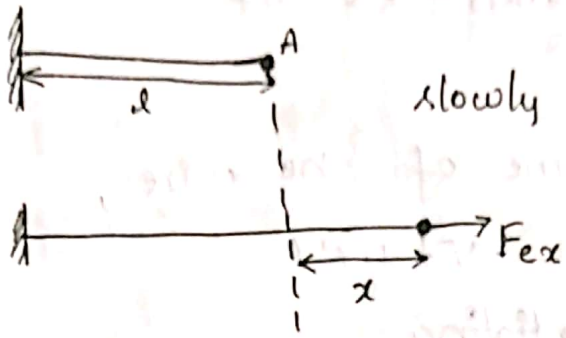
Body is most compressible.

Again, $D = 2r$

$\Delta D = 2\Delta r$ $\quad \frac{\Delta D}{D} = \frac{2\Delta r}{2r} \Rightarrow \boxed{\frac{\Delta D}{D} = \frac{\Delta r}{r}}$

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* Strain Energy (or) Elastic Potential energy Stored in the wire



In eqm,

$$F_{ext} = F_{rest} = \frac{YAx}{l}$$

* work done by F_{ext} during further elongation dx is, $dw = F_{ext} \cdot dx$

* Total work done,

$$w = \int dw = \int_{x=0}^{x=e} \frac{YAx}{l} \cdot dx$$

$$\boxed{w = \frac{YAe^2}{2l}}$$

we have,

$$w_{ext} = U_f - U_i^{\nearrow 0}$$

$$w_{ext} = U_f$$

$\therefore$ strain energy $(U) = \frac{1}{2} \frac{YAe^2}{l}$

$$= \frac{1}{2} \left(\frac{YAe}{l}\right) e$$

$$\boxed{U = \frac{1}{2} Fe}$$

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* strain energy,

$$U = \frac{1}{2} \cdot \frac{F}{A} \cdot \frac{e}{l} \cdot A l$$

$$\boxed{U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{Volume}}$$

(or)

$$U = \frac{1}{2} \times \text{stress} \times \frac{\text{stress}}{Y} \times \text{vol}$$

$$\boxed{U = \frac{1}{2} \times \frac{(\text{stress})^2}{Y} \times \text{Vol}}$$

* strain energy density,

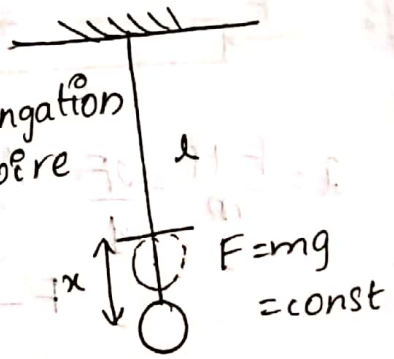
$$u = \frac{U}{\text{Volume}} = \frac{1}{2} \times \text{Stress} \times \text{strain}$$

$$u = \frac{1}{2} \frac{(\text{Stress})^2}{Y} \Rightarrow \boxed{u = \frac{1}{2} (\text{stress})^2 \times \frac{1}{Y}}$$

* In general,

$$U = \int u \, dV = \int \frac{(\text{Stress})^2}{2Y} \, dV$$

Eg: w.r.to initial position,



* change in grav. P.E of

sphere $= -mgx$

* Strain energy produced in wire

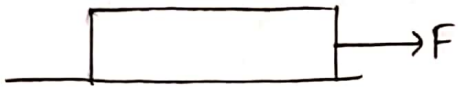
$$= \frac{1}{2} FC = \frac{1}{2} mgx$$

* Heat produced in the wire

$$= \frac{1}{2} mgx //$$

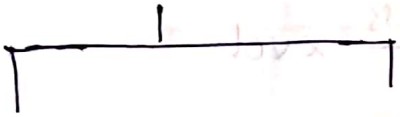
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Eg:



① $a = \frac{F}{m}$ ② $T = F\left(1 - \frac{x}{L}\right)$

③ Stress $= \frac{T}{A} = \frac{F}{A}\left(1 - \frac{x}{L}\right)$



$\frac{de}{dx} = \frac{T}{AY}$ $\quad U = \int u \cdot dv$

$= \frac{1}{2Y} \int_{x=0}^{x=l} \left[\frac{F}{A}\left(1 - \frac{x}{L}\right)\right]^2 A \cdot dx$

$$\boxed{U = \frac{F^2 L}{6AY}}$$

* Relation blw Inter atomic ~~distance~~ force const (K) & young's modulus (Y) is

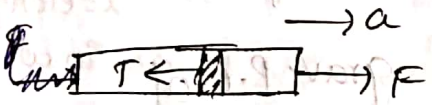
$$\boxed{K = Yr}$$

where r is interatomic distance.

Eg:



$a = \frac{F+F}{m} = \frac{2F}{M}$



$F - T = \frac{m}{l} \times \left(\frac{2F}{m}\right)$

$T = F\left(1 - \frac{2x}{l}\right)$

$S = \frac{T}{A} = \frac{F}{A}\left(1 - \frac{2x}{l}\right)$

$U = \int u \cdot dv = \frac{1}{2Y} \frac{F^2}{A^2} \int_0^L \left(1 - \frac{2x}{l}\right)^2 \cdot A \cdot dx$

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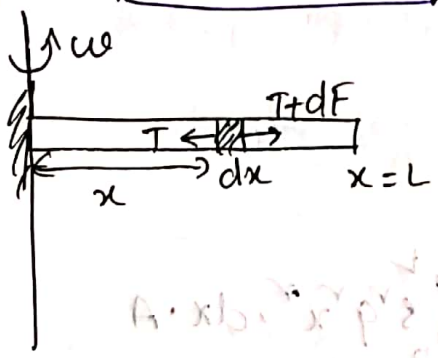
$$U = \frac{1}{2y} \times \frac{F^2}{A} \int_0^L \left[1 + \frac{4x^2}{l^2} - \frac{4x}{l}\right]^2 \cdot dx$$

$$U = \frac{1}{2y} \times \frac{F^2}{A} \left[L + \frac{4L^3}{3l^2} - \frac{4L^2}{2l}\right]$$

$$U = \frac{1}{2y} \times \frac{F^2}{A} \left[\frac{4}{3} - 1\right] L$$

$$\boxed{U = \frac{F^2 L}{6Ay}}$$

Eg 6



$$T - (T + dT) = (dm) \cdot x\omega^2$$

$$-dT = \frac{M}{L} dx\,(x)\omega^2$$

$$T = \frac{m\omega^2}{L}\left(\frac{L^2 - x^2}{2}\right)$$

$$T = \frac{\rho A l \omega^2}{l}\left(\frac{L^2 - x^2}{2}\right)$$

$$T = \frac{1}{2} \rho A \omega^2 (l^2 - x^2)$$

$$S = \frac{1}{2} \rho \omega^2 (l^2 - x^2)$$

$$U = \int u \cdot dv = \frac{1}{2y} \int \frac{\rho^2 \omega^4}{4} (l^2 - x^2)^2 \cdot A \cdot dx$$

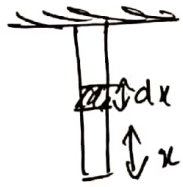
$$U = \frac{\rho^2 \omega^4 A}{2y \times 4} \int_0^l (l^4 + x^4 - 2l^2 x^2) \cdot dx$$

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$$U = \frac{\rho^2 \omega^4 A}{2Y} \times \frac{l^5}{15}$$

$$\boxed{U = \frac{1}{15Y} \times \rho^2 A l \omega^4}$$

Eg:



due to its own wt.

$$T = A\rho g x$$

$$S = \rho g x$$

$$U = \int u \cdot dV = \frac{1}{2Y} \int_0^L \rho^2 g^2 x^2 \cdot dx \cdot A$$

$$U = \frac{A\rho^2 g^2}{2Y} \left[\frac{x^3}{3}\right]_0^L$$

$$\boxed{U = \frac{A\rho^2 g^2 L^3}{6Y}} = \text{heat energy.}$$

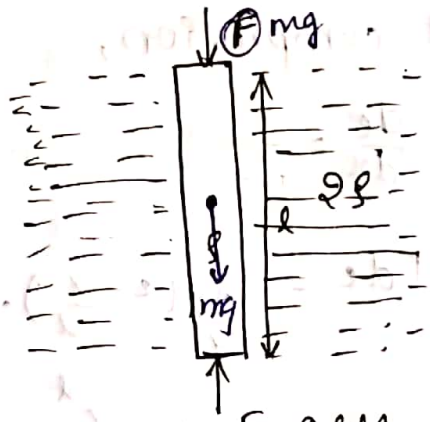
$$\boxed{\text{work done} = 2U}$$

Eg: A thin uniform elastic rod of natural length l, area A, density ρ and young's modulus Y is just completely immersed in vertical position in fluid of density 2ρ by applying a vertically downward force. F at its top end. The rod is in eqm. Find elastic deformation energy stored in the rod (Neglect lateral strain in rod and atm pressure).

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mass of rod

$m = A l \rho$



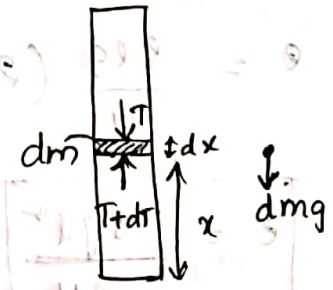
since rod is in eqm,

$F_{net} = 0$

$F + mg = 2mg$

$F = 2\rho g l A = 2mg$

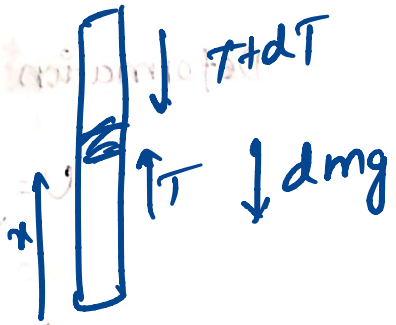
$\boxed{F = mg}$



dT balances dmg.

T changes dx by de.

Both are compressing dx



$-dT = dmg$

$$-\int_{T}^{mg} dT = \int_{x=x}^{x=l} A\rho\, g\, (dx)$$

$T - mg = A\rho g (l - x)$

$T - mg = A\rho g l - A\rho g x$

$T = 2mg - A\rho g x$

$T = 2mg - \frac{mg}{l} x$

$\boxed{T = mg\,(2 - x/l)}$

$T + dT + dmg = T$

$\boxed{dT = -dmg}$

* Stress $= \frac{T}{A} = \frac{mg}{A}\left(2 - \frac{x}{l}\right)$

$\boxed{\frac{T}{A} = l\rho g\,(2 - x/l)}$

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total compression,

$$\frac{de}{dx} = \frac{T}{Ay}.$$

$$de = \frac{\rho l g}{y}\left(2 - \frac{x}{l}\right).dx$$

$$e = \frac{\rho l g}{y}\int_0^l 2 - \frac{x}{l} \cdot dx$$

$$e = \frac{\rho l g}{y}\left(2l - l/2\right)$$

$$\boxed{e = \frac{3\rho g l^2}{2y}}$$

Deformation energy,

$$U = \int u \cdot dv$$

$$U = \frac{1}{2y}\int (\text{Stress})^2 \cdot A \cdot dx$$

$$= \frac{A}{2y}\int \left[l\rho g\left(2 - \frac{x}{l}\right)\right]^2 \cdot dx$$

$$= \frac{A l^2 \rho^2 g^2}{2y}\int_0^l \left[4 + \frac{x^2}{l^2} - 4\frac{x}{l}\right].dx$$

$$= \frac{A l^2 \rho^2 g^2}{2y}\left[4l + \frac{l}{3} - 2l\right]$$

$$= \frac{A l^2 \rho^2 g^2}{2y}\left[\frac{7l}{3}\right]$$

$$\boxed{U = \frac{7A l^3 \rho^2 g^2}{6y}}$$

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M-II &

to find T &-

tension decreases linearly from 2mg to mg.

$T = ax + b$

at $x = 0$; $T = 2mg$

$\boxed{2mg = b}$

& at $x = l$; $T = mg$

$mg = al + 2mg$

$\boxed{a = -\frac{mg}{l}}$

$\therefore T = -\frac{mgx}{l} + 2mg.$

$\boxed{T = mg\left(2 - \frac{x}{l}\right)}$

(OR)

$T = ax + b$

at $x = 0$; $T = mg$

$\boxed{mg = b}$

at $x = l$, $T = 2mg$

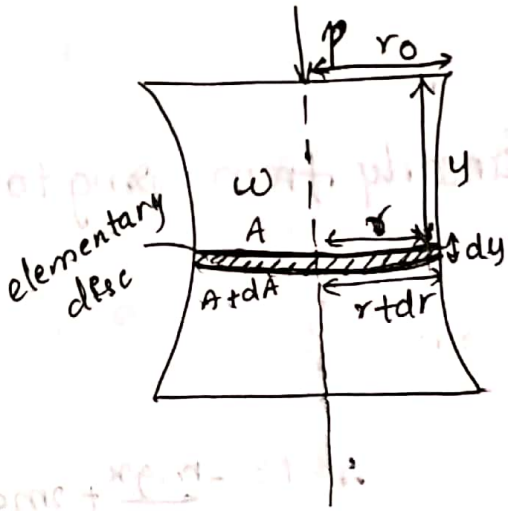
$\Rightarrow \boxed{a = \frac{mg}{l}}$



$\therefore T = \frac{mgx}{l} + mg \Rightarrow \boxed{T = mg\left(1 + \frac{x}{l}\right)}$

Q: A body is in the form of solid revolution, radius of upper base of the body r_0 & volumetric density δ. The solid body supports a load P. determine how the radius should vary with altitude in order that the compressive stress at all cross-sections should be constant.

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here

$$dw = \pi r^2 dy \rho$$

$$dA = \pi(r+dr)^2 - \pi r^2$$

$$= 2\pi r \cdot dr$$

Given, compressive stress

$$\sigma = \frac{P}{\pi r_0^2} = \text{const}$$

i.e, $$\frac{P+w}{A} = \frac{P+w+dw}{A+dA} = \sigma$$

$$PA + wA + PdA + wdA = PA + wA + dw\,A$$

$$(P+w)\,dA = A \cdot dw$$

$$\frac{dw}{dA} = \frac{P+w}{A} = \sigma$$

$$\frac{dw}{dA} = \frac{P}{\pi r_0^2}$$

$$\frac{\pi r^2 dy \rho}{2\pi r \cdot dr} = \frac{P}{\pi r_0^2}$$

$$\int_{r=r_0}^{r=r} \frac{dr}{r} = \int_{y=0}^{y=y} \frac{\pi r_0^2 \rho \cdot dy}{2P}$$

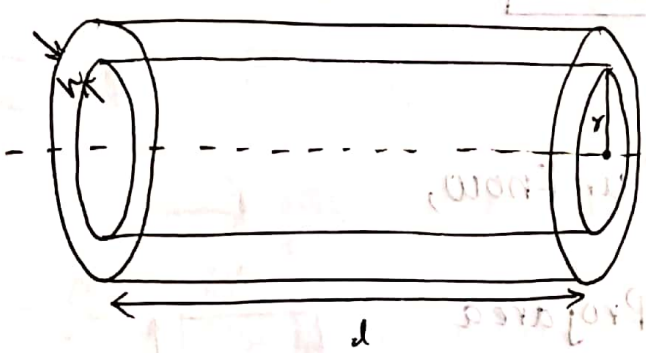
$$\ln(r/r_0) = \frac{\pi r_0^2 \rho y}{2P}$$

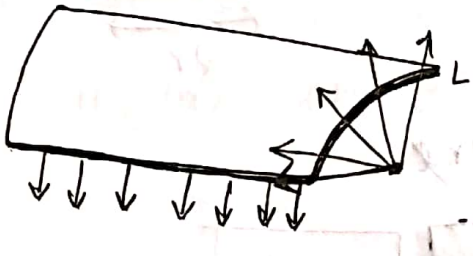
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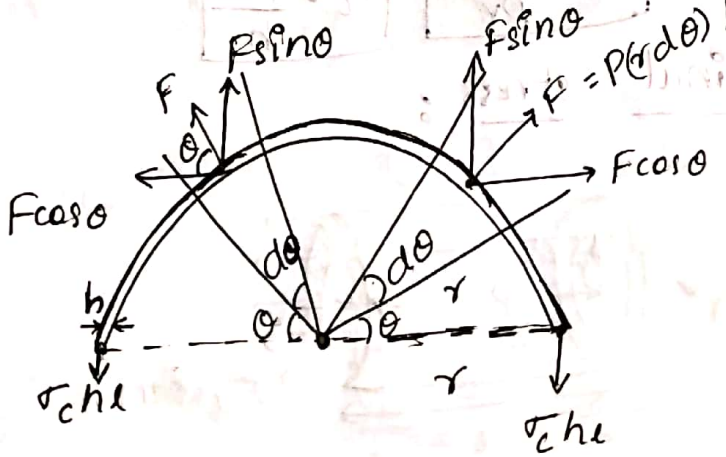
$$\frac{r}{r_0} = e^{\frac{\pi r_0^2 s\gamma}{2P}}$$

$$r = r_0\, e^{\frac{\pi r_0^2 s\gamma}{2P}}$$

Eg: A thin cylindrical shell closed at both ends is subjected to a uniform internal pressure P. The wall thickness is h & the inner radius is r. Calculate the longitudinal & circumferential Normal stresses existing in the walls due to internal pressure.







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In eqm,

$$\Sigma F_V = 0$$

$$2\sigma_c h l = \int_{\theta=0}^{\theta=\pi} F \sin\theta = Pr l \int_0^{\pi} \sin\theta \cdot d\theta$$

$$2\sigma_c h l = Pr l \,(-\cos\theta)_0^{\pi}$$

$$2\sigma_c h = Pr(1+1)$$

$$2\sigma_c h = Pr(2)$$

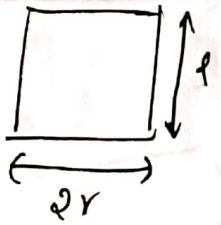
$$\boxed{\sigma_c = \frac{Pr}{h}}$$

M-II:

Since P is unknow,

$$F = P \times \text{Proj area}$$

$$\boxed{F = P 2rl}$$

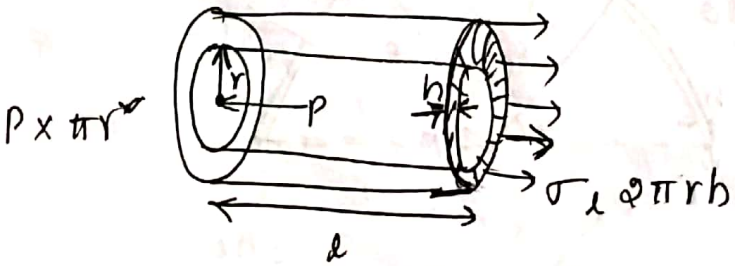


In eqm,

$$2\sigma_c h l = P 2rl$$

$$\boxed{\sigma_c = \frac{Pr}{h}}$$

* longitudinal stress :



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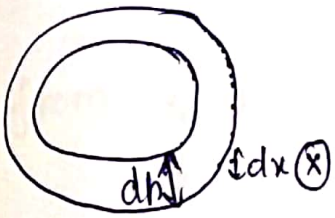
In eqm,

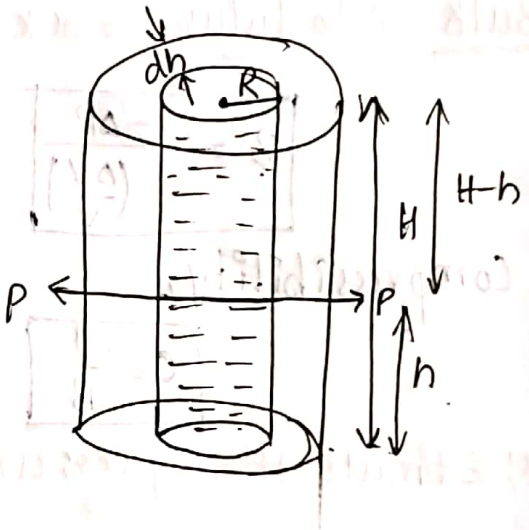
$$P \times \pi r^2 = \sigma_l \times 2\pi r h$$

$$Pr = 2\sigma_l h$$

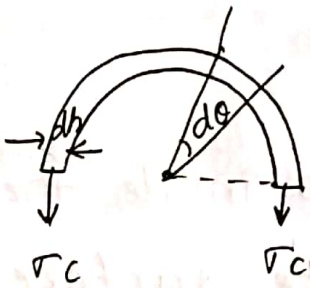
$$\boxed{\sigma_l = \frac{Pr}{2h}} \quad \Rightarrow \sigma_c > \sigma_l$$

Eg: A cylindrical vessel of height H & radius R contains liquid of density s. determine the circumferential tension at a height h. Also determine its max & min values.





$\Rightarrow$



$$\sigma_c \, dh \, (dx) = PR \, (dx)$$

$$\sigma_c = \frac{PR}{dh}$$

$$\frac{F}{(dx)(dh)} = \frac{PR}{dh}$$

$$\frac{F}{dx} = PR$$

↓ Circumferential Tension

$$\Rightarrow \boxed{T = RP}$$

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$T = PR$

$$\boxed{T = \rho g (H-h) R}$$

when $h = 0 \Rightarrow T = T_{max}$

$\therefore T = T_{max}$ (at bottom)

when $h = H$,

$$\boxed{T = T_{min} = 0}$$ (at top)

Bulk Modulus (B or K) :-

$$\boxed{B = \frac{-\Delta P}{\left(\frac{\Delta V}{V}\right)}}$$

Compressibility,

$$\boxed{C = \frac{1}{B}}$$

Ex: Estimate the pressure deep inside the sea at a depth h below the surface. Assume density (ρ) is ρ_0 at sea level & its bulk modulus B.

variation of pressure with depth is,

$$\frac{dp}{dh} = \rho g \quad -①$$

Bulk modulus,

$$B = \frac{-dp}{\frac{dv}{v}} \quad -②$$

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vol of the fluid at depth 'h' is,

$$V = \frac{M}{\rho}$$

on differentiating,

$$dV = \frac{M}{\rho^2} \cdot d\rho$$

$$\frac{dV}{V} = \frac{-M}{\rho^2} \cdot \frac{d\rho}{M} \rho$$

$$-\frac{dV}{V} = \frac{d\rho}{\rho} \quad -③$$

from eqns ①, ②, ③

$$\frac{dP}{B} = \frac{d\rho}{\rho} \quad -④$$

⇒ and $\frac{\rho g dh}{B} = \frac{d\rho}{\rho}$

$$\int_0^h \frac{g\,dh}{B} = \int_{\rho_0}^{\rho} \frac{d\rho}{\rho}$$

$$\frac{gh}{B} = \left[-\frac{1}{\rho}\right]_{\rho_0}^{\rho}$$

$$\frac{gh}{B} = -(1/\rho - 1/\rho_0)$$

$$\frac{gh}{B} = \frac{1}{\rho_0} - \frac{1}{\rho}$$

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$$\frac{gh}{B}\rho_0 = 1 - \frac{\rho_0}{\rho}$$

$$\frac{\rho_0}{\rho} = 1 - \frac{gh\rho_0}{B} \quad -(5)$$

from eqn (4)

$$\int_{P_0}^{P} dp = B\int_{\rho_0}^{\rho} \frac{d\rho}{\rho}$$

$$P - P_0 = B \ln(\rho/\rho_0)$$

$$P = P_0 + B \ln(\rho/\rho_0)$$

$$P = P_0 - B \ln(\rho_0/\rho)$$

$$\boxed{P = P_0 - B \ln\left(1 - \frac{\rho_0 gh}{B}\right)}$$

Important formala &

Energy density of fluid,

$$u = \frac{(\text{stress})^n}{2y} \Rightarrow \boxed{u = \frac{p^n}{2y}} \quad (\text{stress} = p)$$

∴ energy stored in the fluid,

$$\boxed{U = \int u \cdot dv}$$

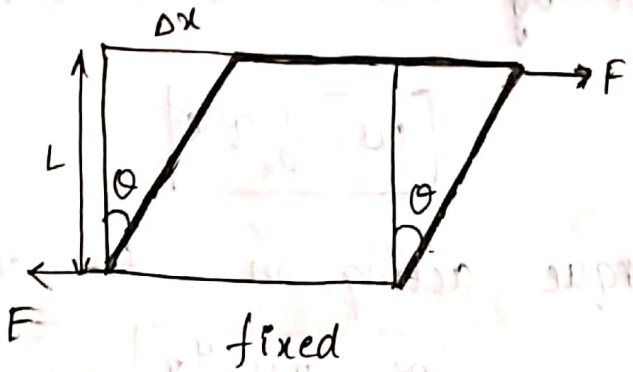
Rigidity Modulus (or) shear modulus (η) &

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Angle of shear (or) strain

$\theta = \frac{\Delta x}{L}$

$\Rightarrow \boxed{\eta = \frac{F}{A\theta}}$



Eg: punching a hole in the block:

$\frac{F}{A} \geq$ ultimate shearing stress

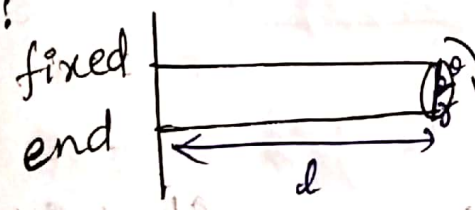
$F_{min} = [\text{ults hearing . stress}] \times \text{Area}$

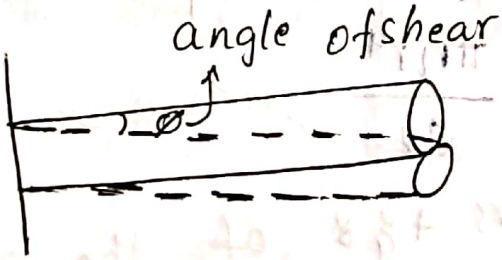
↓

$(2\pi rt)$

↳ curved surface area

Eg:





from figure,

$S = r\theta = l\phi$

$\boxed{\phi = \frac{r\theta}{l}}$

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* working done in twisting the wire,

$$\boxed{W = \frac{1}{2} \tau \theta}$$

* Torque acting on the cylinder is,

$$\boxed{\tau = \frac{\pi \eta r^4 \theta}{2l}}$$ (for solid cylinder)

$$\boxed{\tau = \frac{\pi \eta (r_2^4 - r_1^4) \theta}{2l}}$$ (for hollow cylinder)

* $\therefore W = \frac{1}{2} \times \frac{\pi \eta r^4 \theta^2}{2l}$

$$\boxed{W = \frac{\pi \eta r^4 \theta^2}{4l}}$$

$$\boxed{W = \frac{1}{2} C \theta^2}$$

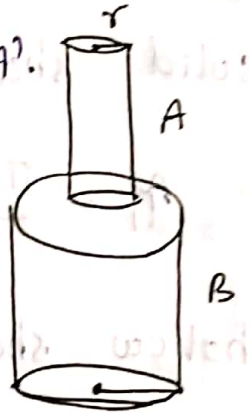
C = restoring torque per unit twist

$$= \frac{\pi \eta r^4}{2l}$$

Eg: Two cylinders A & B of the same material have same lengths, their radii being in the ratio 1:2 respectively they are joined end to end as shown.

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The upper end A is rigidly fixed. The lower end of B is twisted through an angle θ. Find the angle of twist of the cylinder A.



Let θ_1 = angle of twist of 'A'.

angle of twist of B is $\theta - \theta_1$.

Here $\tau_1 = \tau_2$

$$\frac{\pi r^4 \theta_1}{2l} = \frac{\pi r (2r)^4 (\theta - \theta_1)}{2l}$$

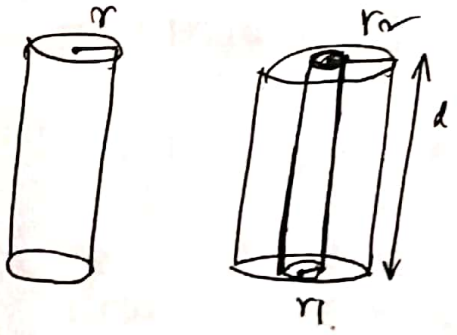
$$\theta_1 = 16(\theta - \theta_1)$$

$$17\theta_1 = 16\theta$$

$$\boxed{\theta_1 = \frac{16\theta}{17}}$$

for B $= \theta - \frac{16\theta}{17} = \frac{\theta}{17}$.

Eg: Show that a hollow shaft is stronger than solid shaft made of same material and of equal volume.



Vol $V = \pi r^2 l = \pi (r_2^2 - r_1^2) l$

$\Rightarrow r^2 = r_2^2 - r_1^2$ — ①

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Torque per unit twist ($\theta = 1$)

$$\tau = \frac{\pi \eta r^4}{2l}$$

for solid shaft;

$$\tau_1 = \frac{\pi \eta \cdot r^4}{2l}$$

for hollow shaft,

$$\tau_2 = \frac{\pi \eta (r_2^4 - r_1^4)}{2l}$$

$$\frac{\tau_1}{\tau_2} = \frac{r^4}{r_2^4 - r_1^4} = \frac{r^2 \cdot r^2}{r_2^4 - r_1^4}$$

$$\frac{\tau_1}{\tau_2} = \frac{(r_2^2 - r_1^2)(r_2^2 - r_1^2)}{(r_2^2 - r_1^2)(r_2^2 + r_1^2)} < 1$$

$$\boxed{\tau_1 < \tau_2}$$

** Relation b/w Y, η, k & σ

① $Y = 3k(1 - 2\sigma)$ ✓

② $Y = 2n(1 + \sigma)$ ✓

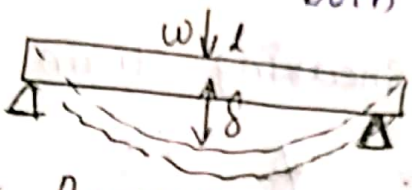
Bending of Beams:

Beam is streched member which can carry transverse load.

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Types of beams :

① Simply supported beam :- which is supported at its both ends.



② Cantilever Beam :- which is fixed at one end.



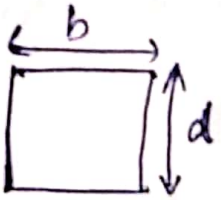
Deflection of beam for simply supported beam — $$\delta = \frac{wl^3}{48EI}$$

Deflection of beam for cantilever beam-

$$\delta = \frac{wl^4}{3EI}$$

where I = geometric moment of area.

① for rectangular cross-section.



$$I = \frac{bd^3}{12}$$

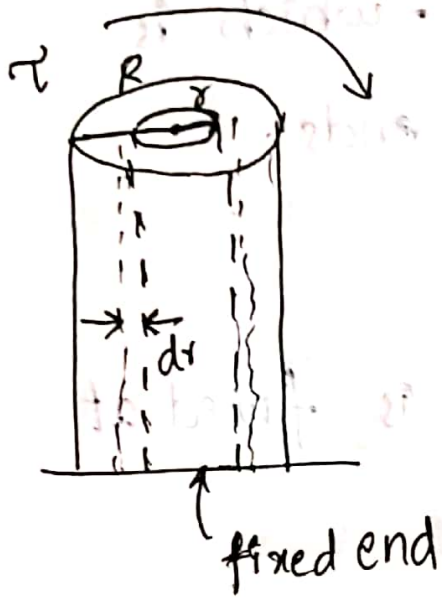
② for circular cross-section



$$I = \frac{\pi r^4}{4}$$

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Torque required in twisting the wire :-



θ = angle of twist

Shearing strain of element,

$$\phi = \frac{r\theta}{l}$$

Stress on small element $= \eta\phi$

$$= \frac{\eta r \theta}{l}$$

Torque per unit area in twisting the element

$$= \text{Stress} \times r = \frac{\eta r^2 \theta}{l}$$

Torque required in twisting small area,

$$d\tau = \frac{\eta r^2 \theta}{l} \times 2\pi r\, dr$$

Total torque required, $\tau = \int d\tau = \frac{2\pi\theta}{l} \int_0^R r^3 \, dr$

$$\boxed{\tau = \frac{\pi \eta R^4 \theta}{2l}}$$

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eg: Sea water has bulk modulus of $23 \times 10^{10} N/m^2$. Calculate the s of sea water at a depth where the pressure is 800atm. If the density at the surface is $1024 kg/m^3$.

We have,

$$B = -\frac{\Delta P}{\frac{\Delta V}{V}}$$

$$B = \frac{+\Delta P}{+\frac{\Delta \rho}{\rho}} \quad \left(\because \frac{\Delta V}{V} = -\frac{\Delta \rho}{\rho}\right)$$

$$\frac{\Delta \rho}{\rho} = \frac{\Delta P}{B}$$

$$\frac{\rho_1 - \rho}{\rho} = \frac{\Delta P}{B}$$

$$\frac{\rho_1}{\rho} = 1 + \frac{\Delta P}{B}$$

$$\rho_1 = \rho\left(1 + \frac{\Delta P}{B}\right)$$

$$= 1024\left(1 + \frac{799 \times 10^5}{23 \times 10^{10}}\right)$$

$$= 1024.4 \ kg/m^3$$

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Eg: Find the depth of the lake at which the density is 1% greater than at the surface. Given compressibility is 50×10^{-6}/atm

we have,

$$\frac{\Delta\rho}{\rho} = \frac{\Delta P}{B}$$

$$\frac{1}{100} = \frac{h\rho g}{B}$$

$$\frac{1}{100} = \frac{h\times10^{3}\times10}{B}$$

$$\frac{1}{100} = \frac{h\times10^{4}\times50\times10^{-6}}{10^{5}}$$

$$h = \frac{1}{50}\times10^{4}$$

$$h = \frac{10}{5}\times10^{3}\,m$$

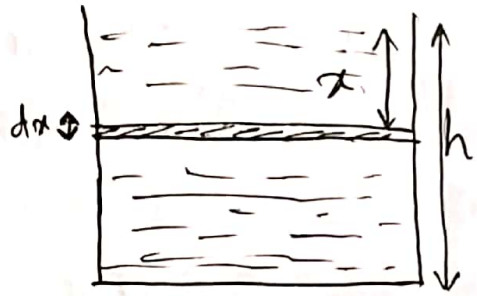
$$\boxed{h = 2\,km}$$

Eg: In a vertical cylindrical vessel of base area $80cm^2$ water is filled to a height $h = 30cm$. If density & bulk modulus of water $\rho = 10^3\,kg/m^3$ & $B = 2\times10^9\,N/m^2$. take $g = 10\,m/s^2$. Find elastic deformation energy of the water in the vessel.

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$$U = \int u \cdot dv$$

$$= \int_{x=0}^{x=h} \frac{p^2}{2B} \cdot A \cdot dx$$



$$= \frac{s^2 g^2 A}{2B} \int_0^h x^2 \cdot dx \qquad P = s g x$$

$$= \frac{s^2 g^2 A}{2B} \frac{h^3}{3}$$

$$U = \frac{10^6 \times 10^2 \times 80 \times 10^{-4}}{2 \times 2 \times 10^9} \times \frac{9 \times 10^3 \times 10^{-4} \times 2}{3}$$

$$U = 18 \times 10^{-7}$$

$$\boxed{U = 1.8 \times 10^{-6}}$$

$$U = \frac{10^6 \times 10^2 \times 80 \times 10^{-4}}{2 \times 2 \times 10^9 / 3} \times \frac{9 \times 10^3 \times 10^{-6}}{3}$$

$$U = 18 \times 10^{-7}$$

$$\boxed{U = 1.8 \times 10^{-6}\ J}$$

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